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JOMHACK
EMPOWERING INNOVATION

JomCode Coding Bootcamp

CHAPTER 19

JavaScript values, functions and modules

learn / type / run / explain

Plain meaning and technical view

PLAIN MEANING

Browser logic uses values and reusable functions just like Python, with different syntax.

TECHNICAL VIEW

const/let, strict equality, branches, loops, arrow functions and ESM imports define JavaScript behavior.

REAL-WORLD USE

Validate form data and share helpers before using React.

QUICK TIP

Use === for comparison. const stops reassignment of a binding but does not freeze an object.

INSTALL / RUN

Node is installed. Run node FILE.js from workbooks. domain.js is supplied.

Worked example 1: Types and conversion

chapter_19_javascript_values_functions_and_modules.js

```
01  const countText = "12";
02  const count = Number(countText);
03  const image = null;
04  console.log(typeof countText, typeof count);
05  console.log(count + 8, image === null);
06  console.log(Number.isNaN(Number("many")));
```

PREDICT Why does conversion need a NaN check?

Run and explain example 1

EXPECTED OUTPUT

```
string number
20 true
true
```

WHY IT BEHAVES THIS WAY

Invalid numeric text produces NaN. JavaScript null represents intentional absence.

Worked example 2: Branches and a loop

chapter_19_javascript_values_functions_and_modules.js

```
01  const titles = ["", "Hello", "A".repeat(161)];
02  for (const title of titles) {
03    if (title.trim().length === 0) console.log("blank");
04    else if (title.length > 160) console.log("too long");
05    else console.log("valid");
06  }
```

PREDICT How do braces compare with Python indentation?

Run and explain example 2

EXPECTED OUTPUT

```
blank
valid
too long
```

WHY IT BEHAVES THIS WAY

Braces define the branch body. const in a loop creates a binding for each iteration.

Exercise 1: Reusable validation

YOUR TASK

Write a function returning valid/blank/too long for a trimmed title. Run all three cases.

CHECK THESE CASES

Spaces blank. Hello valid. 161 characters too long.

HINT AFTER A TRY

Return a value instead of printing inside the helper.

Worked example 3: Functions and return

chapter_19_javascript_values_functions_and_modules.js

```
01  function preview(title, length = 5) {
02    return title.trim().slice(0, length);
03  }
04  const normalize = (text) => text.trim().toLowerCase();
05  console.log(preview(" Python "));
06  console.log(normalize(" ADA "));
```

PREDICT What happens if return is omitted in the function body?

Run and explain example 3

EXPECTED OUTPUT

```
Pytho
ada
```

WHY IT BEHAVES THIS WAY

The caller receives undefined. An arrow expression can implicitly return its expression.

Worked example 4: Named module import

chapter_19_javascript_values_functions_and_modules.js

```
01  import { cleanTitle } from "./domain.js";
02  console.log(cleanTitle(" Hello "));
```

PREDICT Why omit a global variable here?

Run and explain example 4

EXPECTED OUTPUT

```
Hello
```

WHY IT BEHAVES THIS WAY

The exported helper has an explicit name and import path. In browser modules use a local server.

Exercise 2: Module and preview

YOUR TASK

Reuse the supplied cleanTitle module and build a preview with a default length.

CHECK THESE CASES

Python previews Pytho. Go stays Go. Optional length 2 gives Py.

HINT AFTER A TRY

Imports stay at module top level, outside braces.

Checkpoint before moving on

SHOW AND EXPLAIN

Run all four examples, complete both tasks, and explain validate form data and share helpers before using react.

TRANSFER TO POSTLIGHT

Validate form data and share helpers before using React.